



# Statistics of Financial Markets

## Chapter 1: Data, returns and indicators



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## Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CRSP: Center for Research in Security Prices (University of Chicago)
- ▣ ECB: European Central Bank
- ▣ EOD: End Of Day (daily closing data)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ LSEG: London Stock Exchange Group (data vendor, formerly Refinitiv)
- ▣ MDD: Maximum DrawDown
- ▣ NYSE: New York Stock Exchange
- ▣ OHLCV: Open, High, Low, Close, Volume
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ OUG: Ordonanță de Urgență a Guvernului (government emergency ordinance)
- ▣ SE: Standard Error
- ▣ SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- ▣ SR: Sharpe Ratio
- ▣ US: United States

## Today's Question and Route

- ▣ **Question:** how do we turn raw market data into numbers that let us compare investments?
  - ▶ a price level says little; a return, a risk measure and a drawdown say a lot
- ▣ **Route** of the chapter
  - ▶ data sources and data quality: where prices come from and what can go wrong
  - ▶ simple and log returns, over several periods and for portfolios
  - ▶ annualisation and descriptive statistics of returns
  - ▶ performance indicators: CAGR, volatility, Sharpe, Sortino, maximum drawdown, Calmar, volatility drag
- ▣ Real data throughout: BET, S&P 500, DAX, Bitcoin and five stocks of the BVB

## Learning Outcomes

- ▣ Choose a reliable data source and check a price series before using it
- ▣ Compute simple and log returns, and aggregate them over time and across assets
- ▣ Annualise means and volatilities with the actual frequency of the data
- ▣ Describe a return series by its moments, quantiles and extremes
- ▣ Compute and interpret CAGR, Sharpe, Sortino, maximum drawdown, Calmar and the volatility drag
- ▣ Judge whether a difference between two indicators is larger than its estimation error

## Reading and Tools

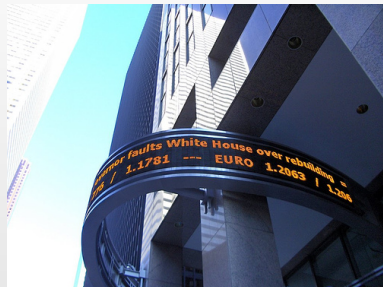
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), *Statistics of Financial Markets*, 5th ed., Ch. 11 (returns, definitions)
  - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#)
- Complementary: [Tsay \(2010\)](#), Ch. 1 (asset returns); [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 1
- Python Quantlets of this chapter: [Quantlets/Ch\\_01](#)
  - ▶ ported from the SFE Quantlets of the textbook (SFEReturns, SFEcomplogreturns, SFEportlogreturns, SFEfdescStat)
  - ▶ each chart in these slides links to the Quantlet that draws it
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Statistics of Financial Markets](#)

## What Financial Data Are

- ▣ **Market data:** what happens on a trading venue
  - ▶ trade prices, best bid and ask quotes, traded volume
  - ▶ index levels, exchange rates, interest rates, crypto prices
- ▣ **Corporate actions:** events that change the price without changing the value of a holding
  - ▶ dividends, splits, share consolidations, bonus shares
- ▣ **Fundamental and macro data**
  - ▶ balance sheets and earnings; inflation, GDP, policy rates
- ▣ **Alternative data**
  - ▶ news, social media, satellite images, blockchain transactions
- ▣ This course works mainly with **daily market data**

## From the Ticker Tape to Data Feeds

- From 1867, **stock tickers** printed trade prices on a paper tape, sent by telegraph
  - ▶ the first real-time price feed: the price of the last trade, minutes after it happened
- Today: electronic feeds from the exchanges, resold by data vendors
  - ▶ every quote and trade, time-stamped to the microsecond
  - ▶ daily summaries (OHLCV) are what most studies use
- The ticker survived: scrolling prices and news on building fronts; ticker symbols such as TLV, SNP, H2O on the BVB



News ticker on the Reuters building, New York

Photo: bgilliard (2005); CC BY-SA 2.0; Wikimedia Commons

## Data Frequency

- ▣ **Tick data:** every trade and quote
  - ▶ millions of records per day; noise from the bid–ask bounce
- ▣ **Intraday bars:** 1, 5 or 60 minutes
  - ▶ used for realised volatility (Chapter 8)
- ▣ **Daily data:** one row per trading day
  - ▶ the standard of this course and of most academic studies
  - ▶ exchanges trade about 250 days a year; crypto markets trade 365 days a year
- ▣ **Weekly, monthly, yearly data**
  - ▶ fewer observations, closer to the horizon of long-term investors
- ▣ Rule: annualise with the **actual** number of observations per year of each series

## Where the Data Come From

| Source                                            | What it provides                                          | Access                     |
|---------------------------------------------------|-----------------------------------------------------------|----------------------------|
| Exchanges (BVB, Deutsche Börse, NYSE)             | official prices, volumes, index levels                    | website, paid feeds        |
| Data vendors (Bloomberg, LSEG, EODHD)             | prices from many venues, adjusted prices, fundamentals    | subscription               |
| Central banks (BNR, ECB, Federal Reserve)         | reference exchange rates, interest rates                  | free                       |
| Statistical databases (FRED, Eurostat)            | macro series, yields                                      | free                       |
| Academic databases (CRSP, Kenneth French library) | cleaned returns, including delisted firms; factor returns | university licence or free |
| Crypto exchanges and aggregators                  | prices 24/7, on-chain data                                | free or subscription       |

- A vendor collects and cleans; the **primary source** of a number (exchange, central bank) is the reference when two sources disagree

## The Data of This Course

- Daily market data from **EODHD** (EOD Historical Data), saved once in the course repository
  - ▶ 91 series: indices, ETFs, stocks (BVB, US, Europe), crypto, exchange rates, bond yields
  - ▶ columns: date, open, high, low, close, adjusted close, volume
- EUR/RON: the official **BNR** reference rate
  - ▶ the EUR/RON series from EODHD has erroneous quotes (next slides)
- Conventions
  - ▶ indices, exchange rates, crypto: *close*; stocks and ETFs: *adjusted close*
  - ▶ weekend quotes and repeated holiday closes are dropped, except for crypto
  - ▶ each series is analysed on its own calendar
- Comparison period of the chapter: 1 January 2015 to 18 September 2026

## Daily Bars: OHLCV

- **OHLCV**: open  $O_t$ , high  $H_t$ , low  $L_t$ , close  $C_t$ , volume  $V_t$  of day  $t$
- By construction  $L_t \leq O_t$ ,  $C_t \leq H_t$ ; the range  $H_t - L_t$  measures the intraday spread of prices
- Banca Transilvania (TLV), the last five trading days in our data:

| Date        | Open  | High  | Low   | Close | Volume    |
|-------------|-------|-------|-------|-------|-----------|
| 14 Sep 2026 | 34.52 | 34.86 | 34.28 | 34.54 | 346 819   |
| 15 Sep 2026 | 34.40 | 34.68 | 33.10 | 34.00 | 676 653   |
| 16 Sep 2026 | 34.00 | 34.10 | 33.20 | 33.40 | 853 078   |
| 17 Sep 2026 | 33.40 | 33.40 | 33.40 | 33.40 | 1 084 380 |
| 18 Sep 2026 | 34.22 | 35.80 | 34.22 | 35.50 | 1 884 988 |

- Check: on 17 September 2026 open = high = low = close with 1 084 380 shares traded; a flat bar with a large volume is suspicious

## Corporate Actions and the Adjusted Close

- The traded close  $P_t$  jumps on the day of a corporate action; the value of a holding does not
- **Adjustment factor**  $A$  on the event day
  - ▶ cash dividend  $D$ :  $A = 1 - D/P_{t-1}$
  - ▶ split  $m$ -for- $n$  ( $m$  new shares for  $n$  old):  $A = n/m$ ; a consolidation is a split with  $m < n$
- **Adjusted close**: every earlier price is multiplied by the factors of all later events,  $P_t^{\text{adj}} = P_t \prod_{k: t_k > t} A_{t_k}$
- Returns computed from adjusted prices include the dividend and have no artificial jumps
- With dividends, the simple return is  $R_t = (P_t + D_t)/P_{t-1} - 1$  [Franke, Härdle & Hafner \(2019\)](#)

## Worked Example: Adjusted Prices

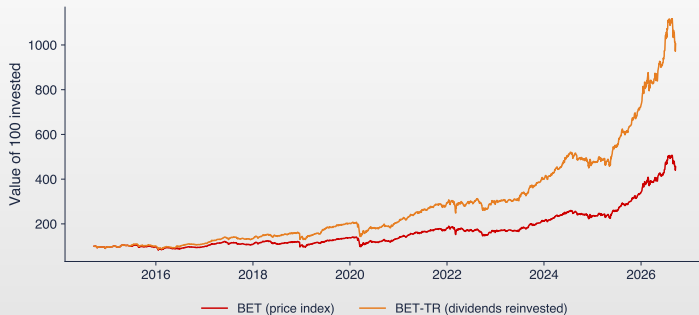
- ▣ A share trades at 100; on day 5 it pays a cash dividend of 2; on day 10 it splits 2-for-1
- ▣ Adjustment factors
  - ▶ dividend:  $A_5 = 1 - 2/100 = 0.98$ ; split:  $A_{10} = 1/2 = 0.50$
- ▣ Adjusted prices
  - ▶ before day 5:  $P_t^{\text{adj}} = P_t \times 0.49$ , so 100 becomes 49
  - ▶ between days 5 and 9:  $P_t^{\text{adj}} = P_t \times 0.50$ ; from day 10:  $P_t^{\text{adj}} = P_t$
- ▣ The adjusted series changes every time a new event happens: always keep the download date
- ▣ Indices: a **price index** ignores dividends; a **total return index** reinvests them

## Real Data: Banca Transilvania, Close vs Adjusted Close



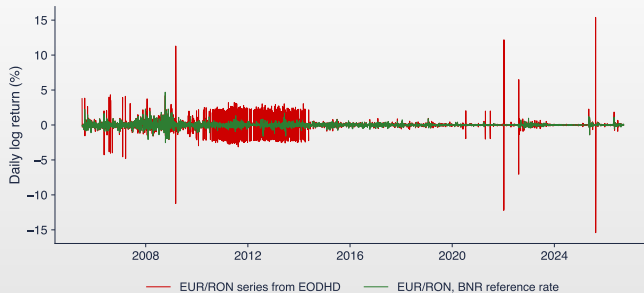
- On 16 August 2022 the close jumps from 2.12 to 21.20 RON: a share consolidation, 10 old shares into one
- Log return from the close: +230% in one day; from the adjusted close: 0.0%
- Since 2015 the two series disagree by more than 2% on 24 days (dividends, bonus shares)

## Dividends Matter: BET vs BET-TR



- 100 invested on 23 September 2014: 455 in the BET (price index), 1 005 in the BET-TR (dividends reinvested)
- Growth rate: 13.5% vs 21.2% a year; dividends add 7.8 percentage points a year
- Compare like with like: the DAX is a total return index, the S&P 500 and the BET are price indices

## Data Errors: Two EUR/RON Series



- Same exchange rate, 5 154 common days: the EODHD series moves by more than 1 percentage point more than the BNR rate on 412 days
- Annualised volatility: 13.4% (EODHD) vs 4.5% (BNR); largest EODHD move 15.4% on 14 August 2025
- Fix: use the primary source of the number, the BNR reference rate

## Three Biases That Make Results Look Too Good

- ▣ **Survivorship bias:** only firms or funds that still exist are in the sample
  - ▶ the losers have disappeared, so the average return is too high [Brown et al. \(1992\)](#)
  - ▶ example: today's members of the BET, followed back to 2000
- ▣ **Look-ahead bias:** using information not yet available at the decision date
  - ▶ annual reports published in March used for decisions in January
- ▣ **Data snooping:** trying many rules on the same data and reporting the best one
  - ▶ hundreds of published return predictors are likely false discoveries [Harvey, Liu & Zhu \(2016\)](#)
- ▣ Each bias inflates returns; none shows up in the summary statistics

## A Data Checklist

- ▣ Source: the primary source of the number and the column used (close or adjusted close)
- ▣ Calendar: weekends, holidays, missing days; same dates for all assets in a joint analysis
- ▣ Jumps: list the 10 largest absolute returns and check each against the news
- ▣ Flat bars, zero volume, repeated prices: holidays filled with the last price or stale quotes
- ▣ Units and currency: RON, EUR, USD; index points; prices in cents
- ▣ Keep the raw file and the download date; never edit data manually

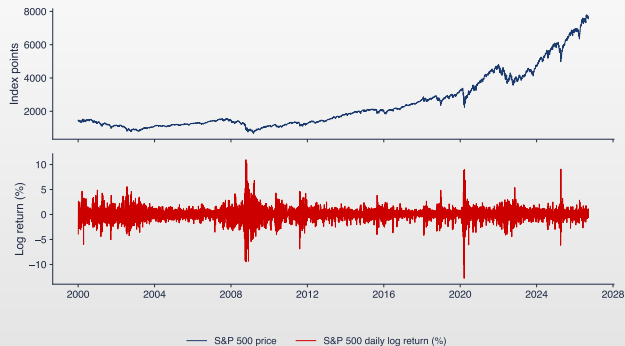
## Recap: Data

- ▣ Prices come from exchanges, vendors and central banks; the primary source of a number is the reference
- ▣ Use adjusted prices for stocks, total return indices when dividends matter
- ▣ Real data contain errors: check jumps, flat bars and a second source
- ▣ Survivorship, look-ahead and data snooping inflate results silently

## Why Returns, Not Prices?

- A price has a unit and a scale; a return does not
  - ▶ a 1-leu move is large for a 2-leu share and small for a 500-leu share
- Returns of different assets, currencies and periods can be compared directly
- Prices trend and wander; returns fluctuate around a stable level
  - ▶ **weakly stationary** series: constant mean, constant variance, and a covariance that depends only on the lag
  - ▶ most statistical methods of this course assume (at least) weak stationarity
- The investor cares about the relative change of wealth, which is exactly a return

## S&P 500: Price Level vs Daily Returns



- Top: the price trends upward with long swings; its mean and spread change over time
- Bottom: the returns fluctuate around zero, in calm and turbulent periods (2008, 2020)

## A Formal Check: the ADF Test

- ▣ **ADF test** Dickey & Fuller (1979): is there a unit root (a random walk) in the series?
  - ▶  $H_0$ : unit root, non-stationary;  $H_1$ : stationary
  - ▶ regression  $\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$ ; reject  $H_0$  when the  $t$ -statistic of  $\gamma$  is very negative
- ▣ S&P 500, daily data since 2000
  - ▶ log price (with trend): statistic  $-2.08$ , 5% critical value  $-3.41$ ,  $p = 0.56$ : we do not reject the unit root
  - ▶ log returns: statistic  $-14.3$ , 5% critical value  $-2.86$ ,  $p < 0.001$ : we reject it
- ▣ Unit roots and random walks are the subject of [Chapter 7](#)

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## The Simple Return

- **Simple (net) return** from  $t - 1$  to  $t$  Franke, Härdle & Hafner (2019), Tsay (2010)
  - ▶  $R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$
  - ▶ **gross return:**  $1 + R_t = P_t / P_{t-1}$
- Bounded below:  $R_t \geq -1$  (limited liability: you cannot lose more than you invested)
- Example:  $P_{t-1} = 100$ ,  $P_t = 110$ :  $R_t = 10\%$
- It is what an investor sees on the account statement: “the fund gained 10%”
- Notation: from here on  $P_t$  is the close of an index or the adjusted close of a stock

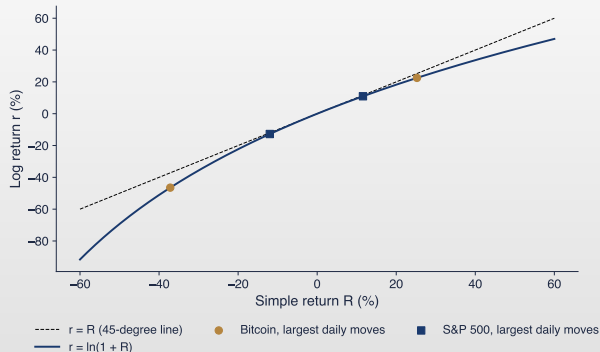
## The Log Return

- **Log return** (continuously compounded return)
  - ▶  $r_t = \ln \frac{P_t}{P_{t-1}} = \ln P_t - \ln P_{t-1} = \ln(1 + R_t)$
  - ▶ back to the simple return:  $R_t = e^{r_t} - 1$
- Name:  $e^r$  is the growth of 1 leu at rate  $r$  compounded continuously over the period
- Unbounded:  $r_t \in (-\infty, +\infty)$ ; a total loss ( $P_t = 0$ ) is  $r_t = -\infty$
- Example:  $P_{t-1} = 100$ ,  $P_t = 110$ :  $r_t = \ln 1.1 = 9.53\%$
- Log returns are the default in statistics of financial markets: they add up over time (next section)

## How Far Apart Are $r$ and $R$ ?

- Taylor expansion of the logarithm around 0
  - ▶  $r = \ln(1 + R) = R - \frac{R^2}{2} + \frac{R^3}{3} - \dots$
  - ▶  $\ln$  is concave, so  $r < R$  for every  $R \neq 0$
- Daily move,  $R = 1\%$ 
  - ▶  $r = 0.995\%$ , and  $R - R^2/2 = 0.995\%$ : the gap  $R - r$  is 0.5 basis points
- Large move,  $R = 10\%$ 
  - ▶  $r = 9.531\%$ , and  $R - R^2/2 = 9.500\%$ : the gap  $R - r$  is 46.9 basis points
- A **basis point** (bp) is 0.01 percentage points
- Conclusion: for daily returns  $r \approx R$ ; for large moves and long horizons they differ

## Simple vs Log Returns on Real Extremes



- Bitcoin, 12 March 2020:  $R = -37.2\%$  but  $r = -46.5\%$
- Bitcoin, 7 December 2017:  $R = +25.2\%$ ,  $r = +22.5\%$
- S&P 500, 16 March 2020:  $R = -12.0\%$ ,  $r = -12.8\%$
- The curve bends down: losses look larger and gains smaller in log returns

## Worked Example: Up 10%, Then Down 10%

- ▣ Price path  $100 \rightarrow 110 \rightarrow 99$ 
  - ▶ simple returns: +10% and -10%; their sum, 0, is misleading
  - ▶ compounded:  $1.1 \times 0.9 - 1 = -1.0\%$
- ▣ Log returns
  - ▶  $r_1 = \ln 1.1 = 9.53\%$ ,  $r_2 = \ln 0.9 = -10.54\%$
  - ▶ sum:  $-1.01\% = \ln(99/100)$ , exact

### Rule

- ▣ Simple returns compound (multiply); log returns add

## Question for the Room: Up 50%, Down 50%

- A share gains 50% one year and loses 50% the next
- **What do you think?** Where is the price after two years, relative to the start?
- **Answer**
  - ▶  $1.5 \times 0.5 = 0.75$ : the wealth changed by  $-25\%$
  - ▶ log returns:  $\ln 1.5 + \ln 0.5 = \ln 0.75 = -28.8\%$
  - ▶ a loss of  $x$  needs a gain of  $x/(1-x)$  to recover: 50% needs 100%

## Which Return When?

| Task                        | Use    | Reason                                        |
|-----------------------------|--------|-----------------------------------------------|
| Aggregating over time       | log    | $r_t(k) = r_t + \dots + r_{t-k+1}$ , a sum    |
| Portfolio of several assets | simple | $R_p = \sum_i w_i R_i$ , exact                |
| Statistical modelling       | log    | unbounded, closer to a symmetric distribution |
| Reporting to investors      | simple | "the fund gained 10%" is a simple return      |
| Daily data, small moves     | either | the difference is about $R^2/2$               |

- Always say which one you use: a "return of 10%" is ambiguous

## Recap: Simple and Log Returns

- ▣  $R_t = P_t/P_{t-1} - 1$ ;  $r_t = \ln(P_t/P_{t-1}) = \ln(1 + R_t)$
- ▣  $r < R$  for every move; the gap is about  $R^2/2$ , negligible for daily data
- ▣ Log returns add over time; simple returns add across assets
- ▣ Gains and losses of the same size do not cancel

## Returns over $k$ Periods

- **Simple:** gross returns multiply

- ▶  $1 + R_t(k) = \frac{P_t}{P_{t-k}} = (1 + R_t)(1 + R_{t-1}) \cdots (1 + R_{t-k+1})$

- **Log:** returns add

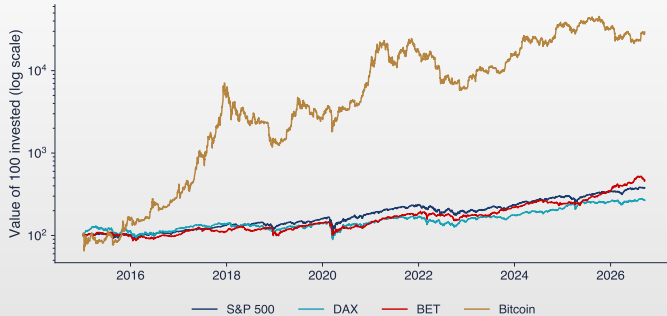
- ▶  $r_t(k) = \ln \frac{P_t}{P_{t-k}} = r_t + r_{t-1} + \cdots + r_{t-k+1}$

- Consequence: the mean and variance of a  $k$ -day log return follow from those of the daily returns
- Weekly, monthly or yearly returns are computed from the prices at the end of each period, not by averaging

## Cumulative Return and Wealth

- Wealth after  $T$  periods, starting from  $W_0$ :  $W_T = W_0 \prod_{t=1}^T (1 + R_t) = W_0 \exp\left(\sum_{t=1}^T r_t\right)$
- **Cumulative simple return:**  $CSR_T = W_T/W_0 - 1$ 
  - ▶ what the investor gained, in percent
- **Cumulative log return:**  $CLR_T = \ln(W_T/W_0) = \sum_t r_t$ 
  - ▶ link:  $CSR_T = e^{CLR_T} - 1$
- Log scale on charts: equal vertical distances are equal percentage changes

## Growth of 100 Invested, 2015–2026



- Final values: S&P 500 379, DAX 267, BET 470, Bitcoin 29 475
- Bitcoin: cumulative simple return 29 375%, cumulative log return 5.69; BET: 370% and 1.55
- Without the log scale, every series except Bitcoin would look flat

## Portfolio Returns

- Portfolio with weights  $w_1, \dots, w_n$ ,  $\sum_i w_i = 1$ , set at the start of the period
  - ▶ **simple:**  $R_p = \sum_i w_i R_i$ , exact
  - ▶ **log:**  $r_p = \ln(\sum_i w_i e^{r_i}) \neq \sum_i w_i r_i$
- Example: 60% in A ( $R_A = 10\%$ ), 40% in B ( $R_B = -5\%$ )
  - ▶  $R_p = 0.6 \times 10\% + 0.4 \times (-5\%) = 4.0\%$ ; exact  $r_p = \ln(1 + R_p) = 3.92\%$
  - ▶ weighted log returns:  $0.6 \times 9.53\% + 0.4 \times (-5.13\%) = 3.67\%$ , too low
- Weights drift when prices move: a fixed-weight portfolio must be **rebalanced**

## Real Data: a TLV–SNP Portfolio

- ▣ 50% Banca Transilvania, 50% OMV Petrom, rebalanced daily, 2015–2026
- ▣ One day: exact log return vs weighted sum of log returns
  - ▶ mean absolute gap 0.38 basis points; largest gap 13.9 basis points, on 31 May 2016
- ▣ Over the whole period the small gaps accumulate
  - ▶ exact cumulative log return 2.56, so 1 leu grows to 13.0 lei
  - ▶ weighted log returns give 2.47, so only 11.8 lei
- ▣ Growth rate: portfolio 24.5% a year; TLV 27.3%; SNP 19.7%
- ▣ Rule: build portfolios with simple returns, then take logs if needed

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## Recap: Multi-Period and Portfolio Returns

- Over time: multiply gross returns or add log returns
- Wealth  $W_T = W_0 e^{\text{CLR}_T}$ ;  $\text{CSR}_T = e^{\text{CLR}_T} - 1$
- Across assets:  $R_p = \sum_i w_i R_i$  exactly; the log version is only an approximation
- Small daily approximation errors become visible over a decade

## Annualising the Mean and the Volatility

- $q$  = number of observations per year: about 252 for exchanges, 365 for crypto, 52 for weekly, 12 for monthly data
- **Mean:** log returns add, so  $\mu_{\text{year}} = q \mu$ 
  - ▶ the same rule is used for the mean of simple returns (arithmetic annualisation)
- **Volatility** = standard deviation of returns:  $\sigma_{\text{year}} = \sqrt{q} \sigma$ , the **square-root-of-time rule**
  - ▶ valid when returns are uncorrelated with constant variance: then  $\text{Var}(r_1 + \dots + r_q) = q \sigma^2$
- S&P 500: daily  $\sigma = 1.12\%$ , so  $\sigma_{\text{year}} = 1.12\% \times 15.87 = 17.7\%$ 
  - ▶ Bitcoin:  $\sqrt{365} = 19.10$  gives 66.6%; the stock-market factor  $\sqrt{252}$  would give only 55.4%

## The Compound Annual Growth Rate

- **CAGR** (compound annual growth rate): the constant yearly rate that turns  $P_0$  into  $P_T$  in  $Y$  years
  - ▶  $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$ , with  $Y$  in calendar years
- Link with log returns:  $\text{CAGR} = \exp(\text{annual mean log return}) - 1$
- S&P 500, 2015–2026: mean log return 11.2% a year,  $e^{11.2\%} - 1 = 11.9\%$ , the CAGR is 11.9%
- The CAGR uses only the first and the last price: it says nothing about the path in between
- Also called the geometric mean return

## How Precisely Do We Know the Mean?

- Standard error of the annual mean:  $SE(\hat{\mu}_{\text{year}}) = \sigma_{\text{year}} / \sqrt{Y}$ , with  $Y$  years of data
  - ▶ it depends on the number of **years**, not on the number of observations
- S&P 500, 11.7 years: annual mean log return 11.2%, SE 5.2%
  - ▶ 95% CI (confidence interval): [1.1%, 21.4%]
- Bitcoin:  $47.4\% \pm 1.96 \times 19.5\%$ 
  - ▶ the interval is [9.2%, 85.6%]: very wide
- Volatility, by contrast, is estimated precisely from daily data

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## Question for the Room: More Data, Better Mean?

- You replace 10 years of daily S&P 500 data by 10 years of 5-minute data: 78 bars in a 6.5-hour trading day, so 78 times more observations
- **What do you think?** Does the standard error of the annual mean return become smaller?
- **Answer**
  - ▶ No: the mean over 10 years is fixed by the first and last price,  $\hat{\mu} = \ln(P_T/P_0)/Y$
  - ▶ only more **years** reduce its standard error, as  $1/\sqrt{Y}$
  - ▶ the volatility estimate, in contrast, does improve with higher frequency

## Recap: Annualisation

- Mean  $\times q$ , volatility  $\times \sqrt{q}$ , with  $q$  the actual observations per year
- $\sqrt{q}$  assumes uncorrelated returns with constant variance
- $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$ , the exponential of the annual mean log return minus 1
- Mean returns are imprecise:  $\text{SE} = \sigma_{\text{year}}/\sqrt{Y}$

## Location and Spread

- ▣ Sample  $r_1, \dots, r_n$  of daily log returns (in %)
- ▣ **Mean:**  $\bar{r} = \frac{1}{n} \sum_{t=1}^n r_t$ , the average daily return
- ▣ **Variance:**  $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2$ ; **standard deviation**  $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ 
  - ▶ in finance the standard deviation of returns is called **volatility**
- ▣ **Quantile**  $q_\alpha$ : the value below which a fraction  $\alpha$  of returns lie
  - ▶ median  $q_{0.5}$ ; the 1% quantile  $q_{0.01}$  is the basis of VaR 1% (Chapter 10)
- ▣ **Minimum** and **maximum:** the worst and best day, always with their dates

## Shape: Skewness and Kurtosis

- **Skewness:**  $\hat{S} = \frac{1}{n} \sum_t (r_t - \bar{r})^3 / \hat{\sigma}^3$ 
  - ▶  $S = 0$ : symmetric;  $S < 0$ : a longer left tail, large losses more frequent than large gains
- **Kurtosis:**  $\hat{K} = \frac{1}{n} \sum_t (r_t - \bar{r})^4 / \hat{\sigma}^4$ ; **excess kurtosis**  $\hat{K} - 3$ 
  - ▶ the Normal distribution has  $K = 3$ , so excess kurtosis 0
  - ▶ excess kurtosis  $> 0$ : **heavy tails**, extreme days more frequent than under the Normal distribution
- Both are sensitive to single extreme days: report them with the extremes
- Heavy tails and asymmetry are **stylised facts** of returns [Cont \(2001\)](#); formal tests in [Chapter 2](#)

## Daily Log Returns: Indices and Bitcoin, 2015–2026

| Series  | Mean  | Std. dev. | Skew. | Exc. kurt. | Minimum (date)      | Maximum |
|---------|-------|-----------|-------|------------|---------------------|---------|
| S&P 500 | 0.045 | 1.12      | −0.64 | 15.8       | −12.8 (16 Mar 2020) | 9.1     |
| DAX     | 0.032 | 1.20      | −0.56 | 10.0       | −13.1 (12 Mar 2020) | 10.4    |
| BET     | 0.053 | 0.98      | −1.41 | 17.8       | −11.9 (19 Dec 2018) | 6.8     |
| BET-TR  | 0.080 | 0.98      | −1.43 | 18.1       | −11.9 (19 Dec 2018) | 6.7     |
| Bitcoin | 0.130 | 3.49      | −0.73 | 12.2       | −46.5 (12 Mar 2020) | 22.5    |

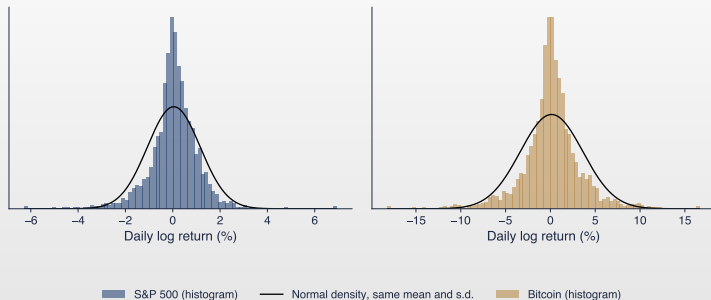
- All values in % except skewness and excess kurtosis; each series on its own calendar
- Bitcoin: daily standard deviation about three times that of the stock indices
- All series: negative skewness and large excess kurtosis
- The worst days cluster in March 2020 (COVID-19) and on 19 December 2018 for the BET

## Daily Log Returns: Five BVB Stocks, 2015–2026

| Series                   | Mean  | Std. dev. | Skew. | Exc. kurt. | Minimum (date)      | Maximum |
|--------------------------|-------|-----------|-------|------------|---------------------|---------|
| Banca Transilvania (TLV) | 0.106 | 1.69      | −0.55 | 25.2       | −22.2 (19 Dec 2018) | 20.5    |
| OMV Petrom (SNP)         | 0.079 | 1.54      | −0.66 | 9.2        | −15.0 (9 Mar 2020)  | 10.0    |
| BRD                      | 0.075 | 1.61      | −0.51 | 9.2        | −18.2 (19 Dec 2018) | 10.3    |
| Nuclearelectrica (SNN)   | 0.118 | 1.53      | 0.29  | 5.2        | −9.4 (9 Mar 2020)   | 11.3    |
| Transgaz (TGN)           | 0.091 | 1.55      | 0.01  | 5.3        | −10.1 (19 Dec 2018) | 11.1    |

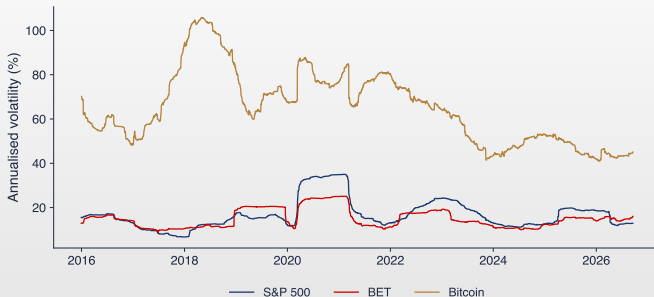
- Single stocks are more volatile than the BET index: diversification at work
- Banca Transilvania and BRD lost most on 19 December 2018, after the announcement of a tax on bank assets, adopted as OUG 114/2018
- Excess kurtosis differs widely between stocks: compare TLV with SNN
- Crashes on the BVB have their own statistics [Pele & Mazurencu-Marinescu \(2012\)](#)

## Histograms vs the Normal Distribution



- Black line: the Normal density with the same mean and standard deviation
- S&P 500: 79.6% of days lie within one standard deviation of the mean (Normal: 68.3%)
- Beyond three standard deviations: 1.39% of days vs 0.27% under the Normal distribution, 5.2 times more

## Volatility Changes over Time



- Rolling one-year volatility, annualised: S&P 500 between 7% and 35%; Bitcoin between 41% and 106%
- One number for the whole sample hides calm and turbulent periods
- Volatility estimators and GARCH models: Chapters 8 and 9

## Recap: Descriptive Statistics

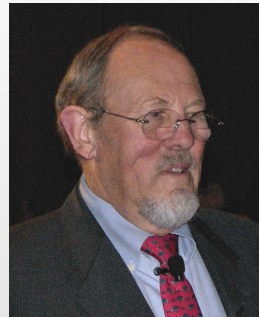
- ▣ Report mean, standard deviation, skewness, excess kurtosis, quantiles and extremes with dates
- ▣ Daily returns: mean near 0, volatility 1–3% a day, heavy tails, often negative skewness
- ▣ Extreme days are several times more frequent than under the Normal distribution
- ▣ Volatility is not constant over time

## Why One Number Is Not Enough

- Return alone rewards risk taking: a leveraged bet can show a high return for years
- Three questions an investor asks
  - ▶ how much did it grow? (CAGR)
  - ▶ how bumpy was the ride? (volatility, downside deviation)
  - ▶ how deep was the worst fall? (maximum drawdown)
- **Risk-adjusted** indicators divide a return by a risk measure: Sharpe, Sortino, Calmar
- All are estimates from one sample: each has an estimation error

## The Sharpe Ratio

- ▣ **Sharpe ratio** Sharpe (1966), Sharpe (1994): excess return per unit of total risk
  - ▶  $SR = \frac{\mu - r_f}{\sigma}$ ;  $\mu$  mean return,  $r_f$  risk-free rate,  $\sigma$  volatility
  - ▶ from daily to annual:  $SR_{\text{year}} = \sqrt{q} SR_{\text{day}}$
- ▣ Here: mean of daily simple returns,  $r_f = 0$ 
  - ▶ the assets are in RON, EUR and USD, each with its own risk-free rate
  - ▶ S&P 500: daily SR 0.046, annual SR 0.72
- ▣ Same Sharpe ratio, same return per unit of risk, whatever the leverage



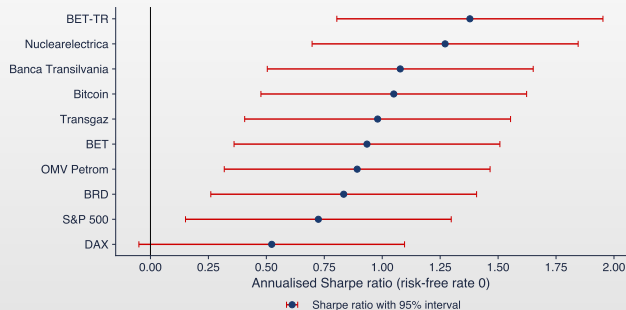
William F. Sharpe, Nobel Prize in  
Economics 1990

Photo: Larry D. Moore (2007); CC BY 4.0;  
[Wikimedia Commons](#)

## The Sharpe Ratio Is an Estimate

- ▣ Standard error for i.i.d. (independent, identically distributed) returns [Lo \(2002\)](#)
  - ▶  $SE(\widehat{SR}) \approx \sqrt{(1 + SR^2/2)/n}$  per period; annual: multiply by  $\sqrt{q}$
  - ▶ roughly  $1/\sqrt{Y}$  for the annual Sharpe ratio: about 0.3 with 12 years of data
- ▣ S&P 500, 2015–2026:  $SR = 0.72$ ,  $SE\ 0.29$ 
  - ▶ 95% CI [0.15, 1.30]
- ▣ Heavy tails and volatility clustering make the true error larger than this formula
- ▣ Choosing the best of many strategies inflates the winner's Sharpe ratio [Bailey & López de Prado \(2014\)](#)

## Sharpe Ratios with 95% Intervals, 2015–2026



- Highest point estimate: BET-TR
- All intervals overlap: the ranking is not statistically significant
- The DAX interval contains 0
- Interval widths are almost equal: they depend mainly on the length of the sample

## The Sortino Ratio

- ▣ Investors dislike losses, not gains: the volatility treats both the same
- ▣ **Downside deviation** below a target  $\tau$  (here  $\tau = 0$ )
  - ▶  $\sigma_D = \sqrt{\frac{1}{n} \sum_t \min(R_t - \tau, 0)^2}$ , only days below the target count
- ▣ **Sortino ratio** [Sortino & van der Meer \(1991\)](#):  $\text{Sortino} = \frac{\mu - \tau}{\sigma_D}$ , annualised like the Sharpe ratio
- ▣ For a symmetric distribution  $\sigma_D \approx \sigma / \sqrt{2}$ , so  $\text{Sortino} \approx 1.4 \times \text{Sharpe}$
- ▣ S&P 500, 2015–2026: Sharpe 0.72, Sortino 1.02

## Drawdown and Maximum Drawdown

- **Drawdown:** the fall from the highest price reached so far

- ▶  $DD_t = \frac{P_t}{\max_{s \leq t} P_s} - 1 \leq 0$ ; zero at a new maximum

- **MDD** (maximum drawdown):  $MDD = \min_t DD_t$ , the worst peak-to-trough fall

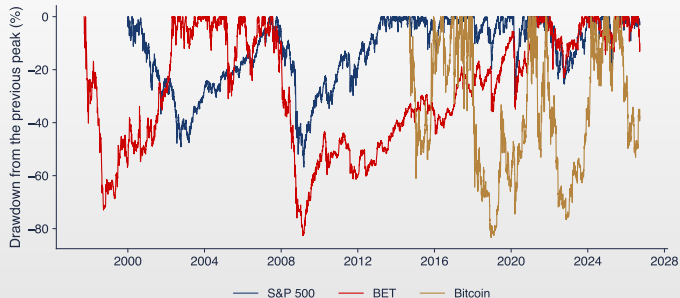
- ▶ also report the dates: peak, trough and **recovery** (back to the old peak)

- A loss  $x$  needs a gain  $x/(1 - x)$  to recover

- ▶ 10% needs 11%; 20% needs 25%; 50% needs 100%; 80% needs 400%

- The MDD depends on the path and grows with the length of the sample [Magdon-Ismail et al. \(2004\)](#)

## Drawdowns over the Full History



- S&P 500: MDD -56.8%, 9 Oct 2007 to 9 Mar 2009, recovered on 28 Mar 2013
- BET (since 1997): MDD -82.5%, 24 Jul 2007 to 25 Feb 2009, recovered only on 16 Mar 2021
- Bitcoin: MDD -83.4%, 16 Dec 2017 to 15 Dec 2018

## Reading the Drawdown Chart

- Depth
  - ▶ after a drawdown of  $-82.5\%$ , the BET needed  $+473\%$  to recover; Bitcoin,  $+502\%$
- Duration
  - ▶ the BET needed 13.6 years to regain its 2007 peak
  - ▶ share of days more than 20% below the peak: S&P 500 27%, BET 56%, Bitcoin 66%
- Volatility and Sharpe ratio ignore the order of returns; the drawdown does not
- In the comparison period 2015–2026, the MDD of each stock index happened in February–March 2020; Bitcoin and several BVB stocks had theirs at other dates

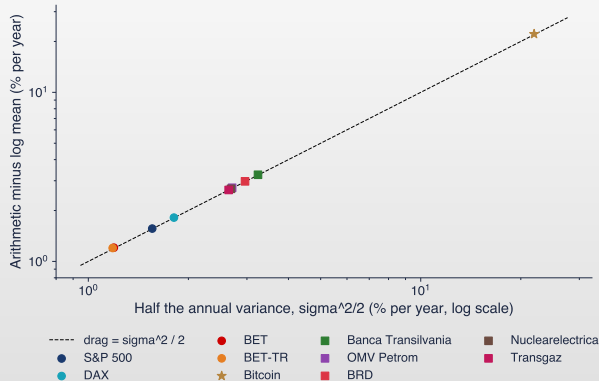
## The Calmar Ratio

- ▣ **Calmar ratio:** growth per unit of the worst fall,  $\text{Calmar} = \frac{\text{CAGR}}{|\text{MDD}|}$
- ▣ Popular with fund managers: the MDD is the loss an investor actually lives through
- ▣ S&P 500, 2015–2026: CAGR 11.9%, MDD –33.9%
  - ▶  $\text{Calmar} = 11.9/33.9 = 0.35$
- ▣ A single event decides the denominator: the Calmar ratio is even noisier than the Sharpe ratio
- ▣ Compare Calmar ratios only over the same period

## Volatility Drag

- For a log return with mean  $m$  and variance  $\sigma^2$ , the simple return has mean  $\approx m + \sigma^2/2$ 
  - ▶ so the growth rate is  $\mu_{\log} \approx \mu_{\text{arith}} - \sigma^2/2$
  - ▶ the gap  $\sigma^2/2$  is the **volatility drag** Booth & Fama (1992); it comes from the concavity of  $\ln$  (Jensen's inequality)
- Two assets with the same arithmetic mean, 10% a year
  - ▶  $\sigma = 5\%$ : drag 0.12%, growth 9.88% a year; after 10 years 1 leu becomes 2.68
  - ▶  $\sigma = 30\%$ : drag 4.50%, growth 5.50% a year; after 10 years 1 leu becomes 1.73
- Doubling the volatility quadruples the drag

## Volatility Drag on Real Data



- Each point: one asset, 2015–2026
- Vertical: arithmetic minus log annual mean; horizontal:  $\sigma^2/2$
- All points lie on the line: the formula works on real data
- Bitcoin: arithmetic mean 69.5%, log mean 47.4%, drag 22.1 percentage points;  $\sigma^2/2 = 21.9\%$

## Question for the Room: Which Bitcoin Return?

- ▣ Bitcoin, 2015–2026: arithmetic mean of simple returns 69.5% a year; CAGR 60.6%
- ▣ **What do you think?** Which number describes the growth of an investor's wealth?
- ▣ **Answer**
  - ▶ the CAGR: it is the rate that actually turned the first price into the last
  - ▶ the arithmetic mean is the expected return of a single year, not the growth over many years
  - ▶ the gap is the volatility drag; for low-volatility assets it is small

## Performance Indicators: Indices and Bitcoin, 2015–2026

|         | CAGR % | Vol. % | Sharpe | Sortino | MDD % | Calmar | Drag % |
|---------|--------|--------|--------|---------|-------|--------|--------|
| S&P 500 | 11.9   | 17.6   | 0.72   | 1.02    | −33.9 | 0.35   | 1.56   |
| DAX     | 8.5    | 19.0   | 0.52   | 0.73    | −38.8 | 0.22   | 1.82   |
| BET     | 14.1   | 15.4   | 0.93   | 1.28    | −31.1 | 0.45   | 1.21   |
| BET-TR  | 22.1   | 15.4   | 1.38   | 1.92    | −31.1 | 0.71   | 1.20   |
| Bitcoin | 60.6   | 66.2   | 1.05   | 1.55    | −83.4 | 0.73   | 22.13  |

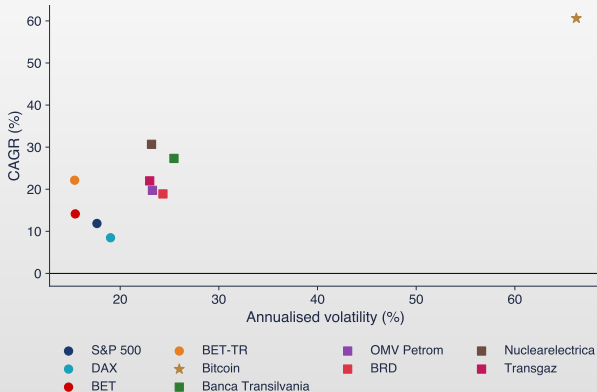
- Annualised with the actual frequency of each series; Sharpe and Sortino with  $r_f = 0$
- Bitcoin: by far the highest CAGR, the highest volatility and the deepest drawdown
- BET-TR vs BET: same risk, dividends lift every return-based indicator

## Performance Indicators: BVB Stocks, 2015–2026

|                          | CAGR % | Vol. % | Sharpe | Sortino | MDD % | Calmar | Drag % |
|--------------------------|--------|--------|--------|---------|-------|--------|--------|
| Banca Transilvania (TLV) | 27.3   | 25.5   | 1.08   | 1.59    | −38.9 | 0.70   | 3.25   |
| OMV Petrom (SNP)         | 19.7   | 23.3   | 0.89   | 1.29    | −45.4 | 0.43   | 2.73   |
| BRD                      | 18.9   | 24.3   | 0.83   | 1.21    | −35.6 | 0.53   | 2.98   |
| Nuclearelectrica (SNN)   | 30.7   | 23.2   | 1.27   | 2.00    | −36.8 | 0.83   | 2.69   |
| Transgaz (TGN)           | 22.0   | 23.0   | 0.98   | 1.49    | −44.8 | 0.49   | 2.65   |

- Adjusted prices: dividends included
- Single stocks: higher volatility and deeper drawdowns than the BET index
- Remember the standard error, about 0.29 for each Sharpe ratio: small differences are not significant

## Risk and Return, 2015–2026



- Horizontal: annualised volatility; vertical: CAGR
- Best by CAGR: Bitcoin; by Sharpe: BET-TR; by Sortino: SNN; by Calmar: SNN
- Lowest volatility: BET-TR; shallowest MDD: BET-TR
- The ranking depends on the indicator: always say which one you use



## Recap: Performance Indicators

- ▣ CAGR (growth), volatility (risk), Sharpe and Sortino (return per unit of risk)
- ▣ MDD and Calmar: the worst fall, which the investor actually lives through
- ▣ Volatility drag:  $\mu_{\log} \approx \mu_{\text{arith}} - \sigma^2/2$ , growth is below the arithmetic mean
- ▣ Every indicator is an estimate: compare differences with their standard errors

## An Open Question

- **Do Sharpe ratios persist?**
  - ▶ if a BVB stock had a high Sharpe ratio in 2015–2020, did it also have one in 2021–2026?
  - ▶ if not, ranking funds or stocks by past Sharpe ratios is of little use to an investor
- Why it is open: short samples, few stocks, and a standard error of about 0.4 for a Sharpe ratio estimated on half of our sample
- Related evidence: many published return patterns do not survive new data [Harvey, Liu & Zhu \(2016\)](#)
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

## How AI Could Help

- ▣ **Literature:** list papers on the persistence of performance and summarise their methods
- ▣ **Code:** draft a Python function for rolling Sharpe ratios on the course data
- ▣ **Robustness:** propose other windows, other indicators (Sortino, Calmar), other markets
- ▣ **Explanation:** turn a table of results into a first draft of the interpretation
- ▣ Example prompt
  - ▶ Write a Python function that computes the annualised Sharpe ratio of daily simple returns on non-overlapping 3-year windows, with the Lo (2002) standard error.

## What to Check

- ▣ Data: adjusted close for stocks, the same calendar for all assets, the right period
- ▣ Formulas: the annualisation factor ( $\sqrt{252}$  vs  $\sqrt{365}$ ), simple vs log returns,  $n$  vs  $n - 1$
- ▣ Numbers: recompute two or three values on small examples or with a second method
- ▣ References: every cited paper must exist; check the DOI
- ▣ Selection: if you tried ten windows, say so; a backtest tried many times overfits [Bailey et al. \(2014\)](#)

## Project Seed

- **Question:** do Sharpe ratios of BVB stocks persist from one period to the next?
  - ▶ data: adjusted closes of 10–15 BVB stocks, 2010–2026, from the course data
- **Steps**
  - ▶ split the sample into two halves; compute Sharpe ratios and their standard errors in each
  - ▶ measure persistence with the rank correlation between the two halves
  - ▶ repeat with Sortino and Calmar ratios
- **Deliverable:** one table, one chart, and a paragraph on what the data can and cannot show
- **Declare any AI use, and list the errors of the AI that you corrected**

## Key Takeaways

- ▣ Check the data first: source, adjusted prices, calendar, jumps
- ▣ Log returns add over time; simple returns add across assets
- ▣ Annualise with the actual frequency:  $\text{mean} \times q$ ,  $\text{volatility} \times \sqrt{q}$
- ▣ Returns have heavy tails, negative skewness and changing volatility
- ▣ Judge performance with several indicators: CAGR, volatility, Sharpe, Sortino, MDD, Calmar
- ▣ Volatility costs growth:  $\sigma^2/2$  a year
- ▣ Mean returns and Sharpe ratios are imprecise: report standard errors

## Key Formulas

| Quantity              | Formula                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------|
| Simple return         | $R_t = P_t/P_{t-1} - 1$                                                                 |
| Log return            | $r_t = \ln(P_t/P_{t-1}) = \ln(1 + R_t)$                                                 |
| $k$ -period returns   | $1 + R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j}), \quad r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$ |
| Portfolio             | $R_p = \sum_i w_i R_i$                                                                  |
| Annualisation         | $\mu_{\text{year}} = q \mu, \quad \sigma_{\text{year}} = \sqrt{q} \sigma$               |
| CAGR                  | $(P_T/P_0)^{1/Y} - 1$                                                                   |
| Sharpe, Sortino       | $(\mu - r_f)/\sigma, \quad (\mu - \tau)/\sigma_D$                                       |
| MDD, Calmar           | $\min_t (P_t / \max_{s \leq t} P_s - 1), \quad \text{CAGR}/ \text{MDD} $                |
| Volatility drag       | $\mu_{\log} \approx \mu_{\text{arith}} - \sigma^2/2$                                    |
| SE of mean, of Sharpe | $\sigma_{\text{year}}/\sqrt{Y}, \quad \sqrt{(1 + \text{SR}^2/2)/n}$                     |

## Check Yourself

- ▣ **Question:** a fund reports daily log returns with mean 0.05% and standard deviation 1%; what are its annual mean and volatility?
  - ▶ **Answer:**  $252 \times 0.05\% = 12.6\%$  and  $\sqrt{252} \times 1\% = 15.9\%$
- ▣ **Question:** why is the arithmetic mean return of Bitcoin far above its CAGR?
  - ▶ **Answer:** the volatility drag  $\sigma^2/2$  is large when  $\sigma$  is large
- ▣ **Question:** an index falls by 60%; what gain does it need to recover?
  - ▶ **Answer:**  $0.6/0.4 = 150\%$
- ▣ Next: [Chapter 2](#), classical distributions and stylised facts of returns

## References (1/2)

- Bailey, D.H., Borwein, J.M., López de Prado, M., Zhu, Q.J. (2014). Pseudo-mathematics and financial charlatanism: the effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, 61(5), 458–471.
- Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- Booth, D.G., Fama, E.F. (1992). Diversification returns and asset contributions. *Financial Analysts Journal*, 48(3), 26–32.
- Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- Brown, S.J., Goetzmann, W., Ibbotson, R.G., Ross, S.A. (1992). Survivorship bias in performance studies. *Review of Financial Studies*, 5(4), 553–580.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Dickey, D.A., Fuller, W.A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Magdon-Ismail, M., Atiya, A.F., Pratap, A., Abu-Mostafa, Y.S. (2004). On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1), 147–161.
- Pele, D.T., Mazurencu-Marinescu, M. (2012). Modelling stock market crashes: the case of Bucharest Stock Exchange. *Procedia – Social and Behavioral Sciences*, 58, 533–542.
- Sharpe, W.F. (1966). Mutual fund performance. *Journal of Business*, 39(S1), 119–138.
- Sharpe, W.F. (1994). The Sharpe ratio. *Journal of Portfolio Management*, 21(1), 49–58.
- Sortino, F.A., van der Meer, R. (1991). Downside risk. *Journal of Portfolio Management*, 17(4), 27–31.

## References (2/2)

- Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- Wang, H., Fu, T., Du, Y., Gao, W., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620, 47–60.